quantum mechanics. Although their predictions always agree completely, the wave function approach and Feynman's sum-over-paths approach give us different ways of thinking about what's going on. As we shall see later on, for some applications, one or the other approach can provide an invaluable explanatory framework.

Quantum Weirdness

By now you should have some sense of the dramatically new way that the universe works according to quantum mechanics. If you have not as yet fallen victim to Bohr's dizziness dictum, the quantum weirdness we now discuss should at least make you feel a bit lightheaded.

Even more so than with the theories of relativity, it is hard to embrace quantum mechanics viscerally—to think like a miniature person born and raised in the microscopic realm. There is, though, one aspect of the theory that can act as a guidepost for your intuition, as it is the hallmark feature that fundamentally differentiates quantum from classical reasoning. It is the uncertainty principle, discovered by the German physicist Werner Heisenberg in 1927.

This principle grows out of an objection that may have occurred to you earlier. We noted that the act of determining the slit through which each electron passes (its position) necessarily disturbs its subsequent motion (its velocity). But just as we can assure ourselves of someone's presence either by gently touching them or by giving them an overzealous slap on the back, why can't we determine the electron's position with an "ever gentler" light source in order to have an ever decreasing impact on its motion? From the standpoint of nineteenth-century physics we can. By using an ever dimmer lamp (and an ever more sensitive light detector) we can have a vanishingly small impact on the electron's motion. But quantum mechanics itself illuminates a flaw in this reasoning. As we turn down the intensity of the light source we now know that we are decreasing the number of photons it emits. Once we get down to emitting individual photons we cannot dim the light any further without actually turning it off. There is a fundamental quantum-mechanical limit to the "gentleness" of our probe. And hence, there is always a minimal disruption that we cause to the electron's velocity through our measurement of its position.